

Individual Diffusion Auralize Display Using an Array of Audio Source Position Tracking Ultrasonic Speakers

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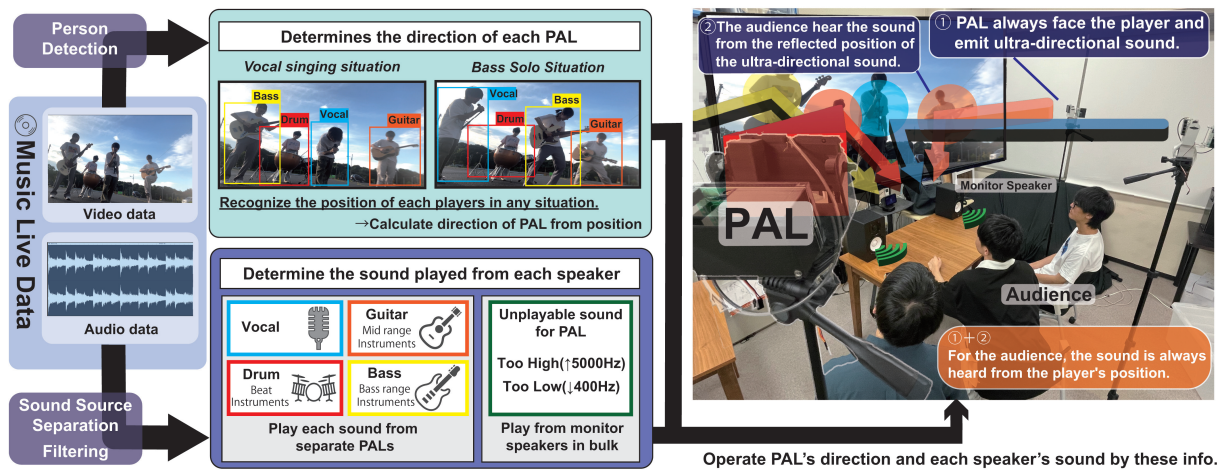


Figure 1: The System Configuration of the Individual Diffusion Auralize Display

Abstract

We developed a prototype system called the "Individual Diffusion Auralize Display," which independently generates each instrument's sound using multiple parametric array loudspeakers (PAL) and monitor speakers. The system adjusts the reflection points of the sounds based on the players' positions. In this study, we demonstrated that the system meets the expected requirements, with the monitor particularly offering an optimal acoustic environment for multi-person viewing.

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1 INTRODUCTION

Spatial audio is crucial in entertainment and interactive experiences like movies, games, and virtual reality. A method for enjoying spatial audio while conversing with multiple people is the surround speaker system. However, the speakers' fixed position and the sound image's localization rely on the balance of volume and phase between the speakers, which leads to issues. These include variations in each individual's head-related transfer function and crosstalk problems, making it challenging to provide a uniform sense of localization for all listeners [Morimoto and Ando 1980].

To address these issues, we focused on parametric array loudspeakers (PAL), which generate ultra-directional sound. It is known that the reflection point of the ultra-directional sound produced by PALs can behave virtually like a sound source [Shi et al. 2014]. We propose a system called the "Individual Diffusion Auralize Display" (Fig. 1), which applies this property to viewing live music videos. This system uses sound source separation to independently reflect

instrument sounds from multiple PALs onto a monitor based on existing live music videos. Furthermore, by tracking the players' positions through person detection, the direction of the PALs is controlled by motors, allowing the reflection positions of each instrument's sound to be adjusted to match the player's position on the display. Additionally, sounds in frequency ranges that are difficult for PALs to emit are output through monitor speakers to maintain the overall acoustic fidelity of the system. This system provides an immersive audio experience, where sound emanates directly from the players on the monitor, independent of the physical placement of the speakers. In this study, we aimed to determine whether the system meets our expected requirements and to identify the most suitable display for it.

2 EXPERIMENTS

In this experiment, we investigated the spatial distribution of the sound pressure levels of the reflected ultra-directional sound generated by PALs and the differences in sound pressure levels depending on the display. The PAL, installed at a height of 1.6 meters from the ground, projected white noise at its maximum sound pressure of 77 dB onto the center of the display (at a height of 1.3 meters). The spatial distribution of the sound pressure levels of the reflected sound was measured by dividing a 2 m × 3 m area in the anechoic chamber into 0.2 m squares and using a sound level meter at each point. The sound level meter was approximately 1.2 meters, considering the ear height when a person is seated, and only the reflected sound was measured. The experimental conditions consisted of four combinations: two types of displays commonly used, "monitor (height 65 cm, width 112 cm)" and "projector screen (polyester material)," and two PAL positions, "in front of the display (150 cm from the left)" and "to the right of the display (94 cm from the left)." Although the actual system envisions using three speakers, adding "to the left of the display (206 cm from the left)," the position "to the left of the display" was assumed to have symmetrical reflection characteristics to the "to the right of the display" position, considering the symmetry of sound reflection.

The evaluation in this experiment was conducted by visualizing the sound pressure distribution of the reflected ultra-directional sound through a sound pressure map. Furthermore, a quantitative evaluation was performed by comparing the horizontal distance d (m) at which the sound pressure exceeded 63 dB from any of the speakers placed to the right, in front of, and to the left (symmetric to the right) of the display at a distance three times the height of the screen from the monitor. The criteria for setting this standard were based on the minimum recommended sound pressure level of 63 dB for music listening [Kim and Han 2018] and the commonly recommended viewing distance for a Full HD television, which is three times the vertical length of the screen [Park et al. 2020]. This evaluation determined whether the system meets the expected requirements and identified the most suitable display for this system.

3 RESULT & DISCUSSION

Figure 2 shows the sound pressure map of the reflected sound measured in the experiment. This map shows that under all conditions,

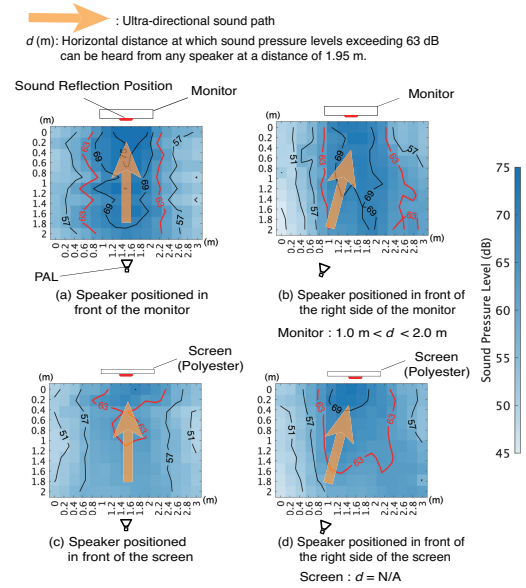


Figure 2: Sound pressure maps of reflected sound.

the reflected sound from the PAL shows the highest sound pressure near the reflection point. While possessing some directivity, the sound diffuses outward from the reflection point. This result indicates that it is possible to make the viewer perceive the sound as originating from the center of the display, which is the reflection point, thus meeting the expected requirements.

Next, we focus on the differences in sound pressure distribution depending on the display used. In the case of the monitor, the vertical height of the screen is approximately 0.65 m. Therefore, when measuring the horizontal distance d at which sound pressure levels exceeding 63 dB could be heard from any speaker at a distance of 1.95 m from the monitor, d was found to range between 1 m and 2 m, ensuring a width of 1 m, which is considered suitable for listening to music by multiple people. On the other hand, assuming the screen height was the same as the monitor's, the sound pressure level at approximately 1.95 m from the screen was below 63 dB under all conditions. Consequently, using a screen for video viewing with this system is considered unsuitable. From these results, it was confirmed that the system meets the expected requirements, and it was found that using a monitor, in particular, provides an optimal acoustic environment for multiple people to view.

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